Course Project: Preliminary Results

DSE6211

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Introduction

To assist ABC Hotels identify bookings with a high risk of cancellation, a preliminary machine learning model has been developed using a dataset of past customer bookings with more than 36,000 records. This report will start with an outline of data processing and encoding steps, then move to an overview of the model with a discussion about model decisions plus a description of the tested model results and will conclude with additional steps to be taken to complete the final report.

Data Processing, Feature Identification, and Data Encoding

As identified in the Analytic Plan document, the dataset was determined to be complete, not missing values and not having unexpected values. There were no records with null values. All but one of the original 17 features will be used as part of this model. One of the features, "arrival_date", will be encoded to "season" to incorporate seasonality into the risk calculation. The dataset of 36,238 records was split into a training dataset of 27,178 records and a test dataset of 9,060 records in a ratio of 75:25.

Below is an updated table for the training dataset that identifies each feature (used or eliminated) and its required data encoding steps.

Column	Data Processing and Encoding Steps
Booking_ID	Eliminated; this is a customer identifier that provides value to the risk calculation
no_of_adults	
no_of_children	
no_of_weekend_nights	
no_of_week_nights	
type_of_meal_plan	OneHotEncoder converted into four features; original feature eliminated
required_car_parking_space	
room_type_reserved	OneHotEncoder converted into seven features; original feature eliminated
lead_time	
arrival_date	Eliminated after converting first to "season" and then to four "season" features via OneHotEncoder
market_segment_type	OneHotEncoder converted into five features; original feature eliminated
repeated_guest	
no_of_previous_cancellations	
no_of_previous_bookings_not_canceled	
avg_price_per_room	
no_of_special_requests	
season	Four features created from "arrival_date" values
booking_status	This is the target label; performed label encoding to turn the categorical values into numeric values. 0 = does not cancel 1 = will cancel

The features were scaled using the StandardScaler from sklearn.preprocessing and were converted to a torch tensor. The training data set was further split into a smaller training set (21,742 records) and a validation set (9,060 records) to be used in the model. The test set was also encoded using OneHotEncoder and StandardScaler.

Model Overview

A feedforward, neural network model was created to perform supervised learning. A Sequential container was used so that each module of the model could be specified. There are two hidden layers that each use PyTorch Linear to apply a linear transformation with weights and biases. These hidden layers each are activated by a ReLU activation function to bring in non-linearity to the model. The final layer uses a Sigmoid function to pull outputs within the 0 to 1 range. (Geeks For Geeks, 2024) The choice to use two hidden layers was made based on the idea of keeping the overall model simple but also layered would capture the complexity of the data. After the initial trial runs, the model seemed to function efficiently.

There are 31 input neurons to the first layer that provided 75 output neurons to the second layer. The next layer returned 50 output neurons to the final layer. The final layer created an output of a single feature, which identified the probability of a customer with similar attributes cancelling their hotel booking. (Navidi, 2020)

The model uses the Adam optimizer (Sahota, 2023) with a learn rate of 1e-3 in an attempt to efficiently achieve convergence. The loss function being used is PyTorch BCELoss as the desired output is binary in nature. (Open Data Science, 2022)

Epoch ranges from 600 to 2000 were tested. A final range of 1000 epochs was chosen because no overfitting was identified and the resulting plot was concise yet informative. Figure 1 displays training and validation loss values.

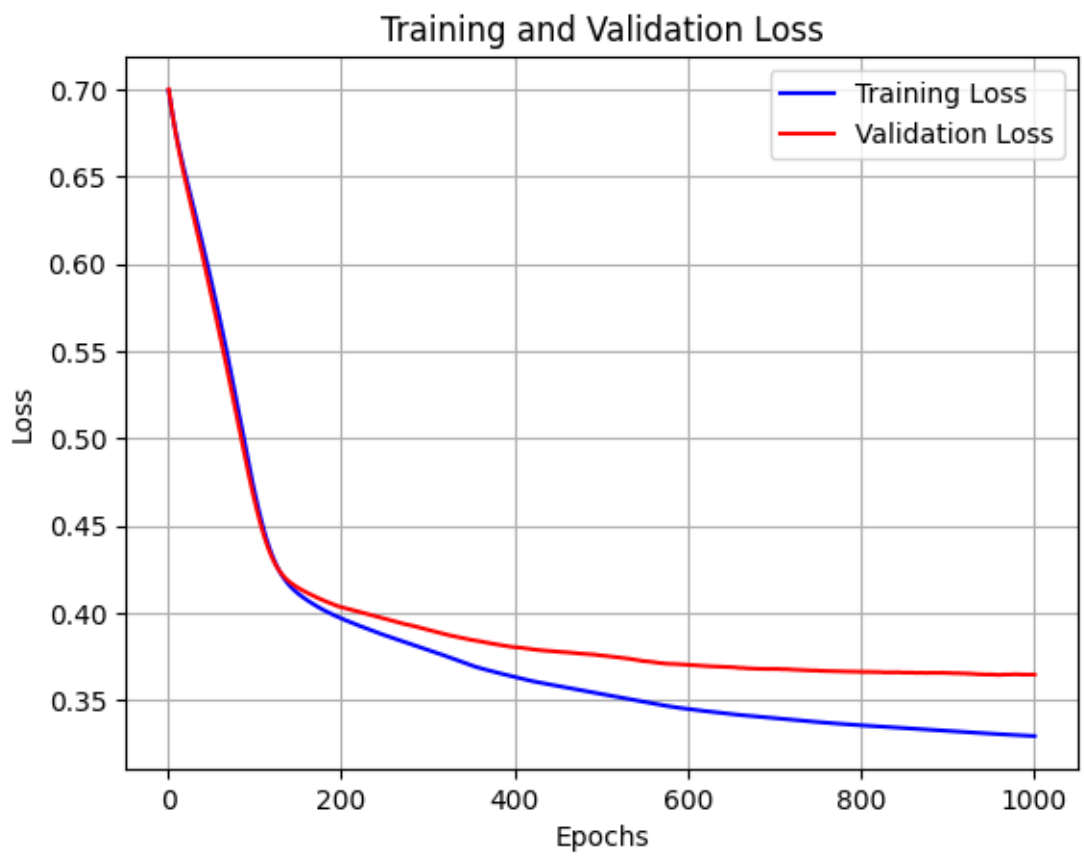

Figure 1: Plots of the training loss and validation loss values

Figure 2 displays the values for the training and validation losses at epoch intervals of 100, up to 1000.

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Epoch 1, Training loss 0.6990, Validation loss 0.7002
Epoch 100, Training loss 0.4695, Validation loss 0.4644
Epoch 200, Training loss 0.3969, Validation loss 0.4034
Epoch 300, Training loss 0.3789, Validation loss 0.3904
Epoch 400, Training loss 0.3634, Validation loss 0.3804
Epoch 500, Training loss 0.3536, Validation loss 0.3757
Epoch 600, Training loss 0.3450, Validation loss 0.3703
Epoch 700, Training loss 0.3396, Validation loss 0.3679
Epoch 800, Training loss 0.3356, Validation loss 0.3663
Epoch 900, Training loss 0.3325, Validation loss 0.3656
Epoch 1000, Training loss 0.3295, Validation loss 0.3646
```

Figure 2 Training and Validation Loss Values

Two additional plots were made to assess how well the model was functioning. Figure 3 is the ROC curve, which displays the curve nearing the upper-left corner of the graph. This suggests that there is a low false positive rate, which is the desired outcome.

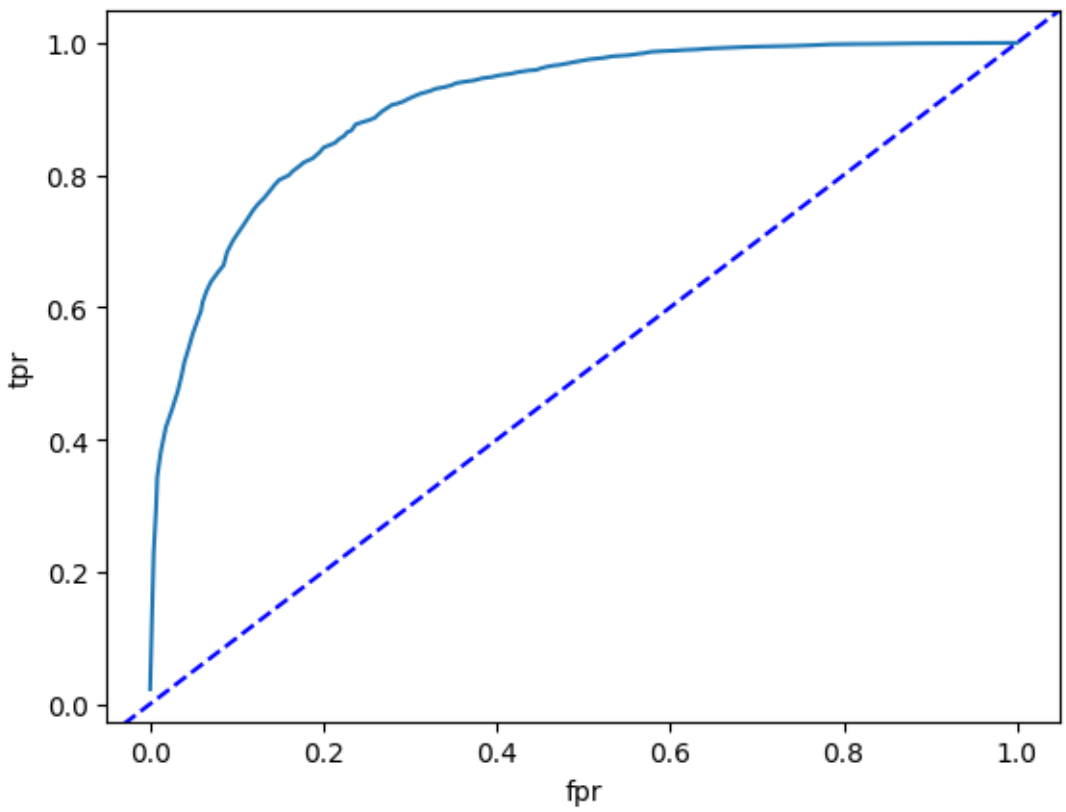

Figure 3: ROC Curve

Figure 4 is a jitter plot that visualizes a balanced class membership, which is also a desired outcome to a binary classification model.

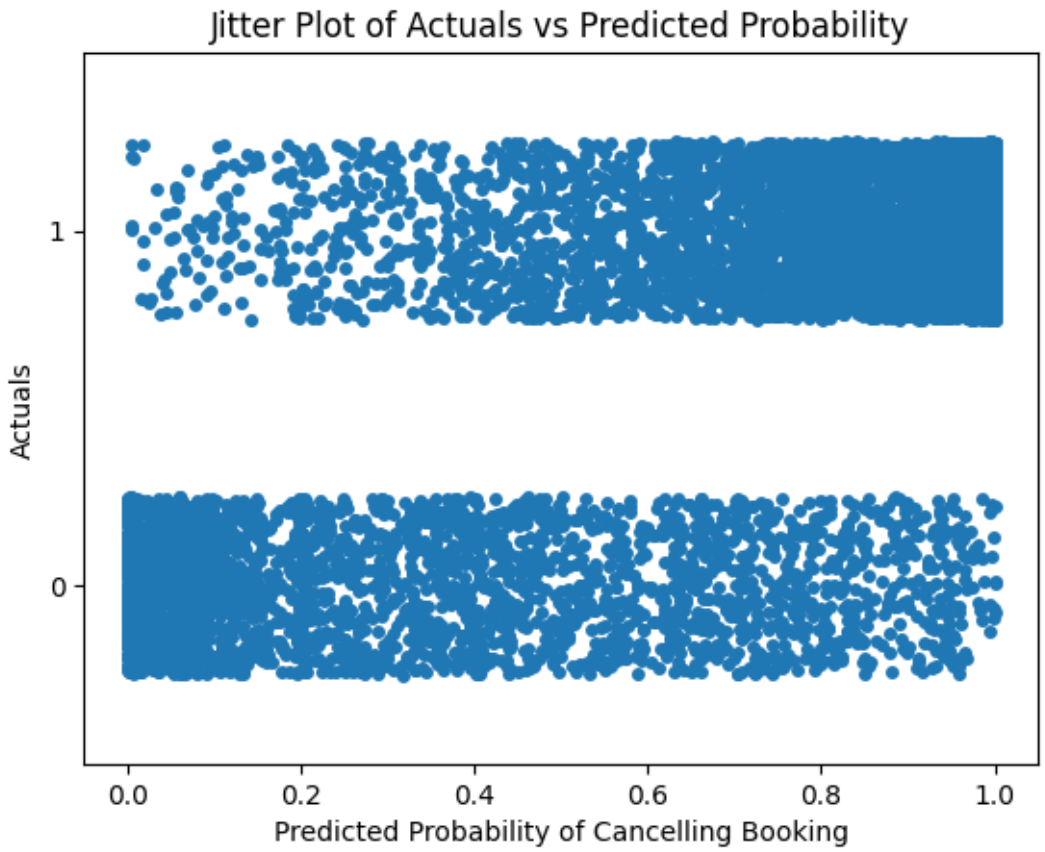

Figure 4 Jitter plot comparison of Actual and Predicted Probability

Conclusion

For the final report, it is desirable to generate a second, slightly different model. The output of this model can be assessed for function and compared to the existing neural network model to support the assertion that the neural network model is effective and efficient. Additionally, adding more methods of evaluation, like accuracy, confusion matrix, Random Guessing and/or calibration curves, would validate the model selection.

Bibliography

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Sahota, H. (2023). *Intuitive Explanation of SGD, Adam, and RMSProp*. Retrieved from Kaggle: https://www.kaggle.com/code/harpdeci/intuitive-explanation-of-sgd-adam-and-rmsprop